UK RAILWAY DATASET

End-to-End Data Analysis Project

From raw CSV to interactive dashboards, Python, SQL, Excel & Power BI

- Final Team Discussion

What this project is about

An end-to-end analysis of a UK railway transactions dataset, moving from raw, inconsistent data to clean insights that support operational and business decisions.

31,653	18	4	0
Ticket transactions	Original columns	Tools used end-to-end	Missing values after cleaning

What we set out to do

1

Assess

Identify and resolve missing values, duplicates and inconsistent categories.

2

Explore

Uncover patterns in journey performance, revenue, delays and passenger behaviour.

3

Enrich

Build metrics, delay time, booking lead days, refund eligibility.

4

Query

Answer business questions in SQL on route, delay and revenue performance.

5

Model

Star schema with Fact + Dimension tables for scalable reporting.

6

Visualise

Interactive Excel dashboard and 3-page Power BI report.

railway.csv — what's inside

Transaction

ID • Date & Time of Purchase • Purchase Type • Payment Method

Ticket

Railcard • Class • Type • Price

Journey

Departure & Arrival Station • Date • Scheduled & Actual Times

Outcome

Journey Status • Reason for Delay • Refund Request

AT A GLANCE

31,653

ticket transactions

18 original columns covering pricing, route, scheduling, delay cause and refund behaviour across multiple UK stations.

Getting to know the data

- **`.shape`**

  Dataset dimensions — 31,653 × 18

- **`.head() / .tail()`**

  Preview first and last rows

- **`.columns`**

  All column names

- **`.dtypes`**

  Data type inspection

- **`.describe()`**

  Summary statistics

Data understanding

- DF shape
- head and tail
- dtypes
- describe

```
[6]: df.shape
[6]: (31653, 18)
[7]: df.head(5)
```

[7]:

	Transaction ID	Date of Purchase	Time of Purchase	Purchase Type	Payment Method	Railcard	Ticket Class	Ticket Type	Price	Departure Station	Arrival Destination	Date of Journey	Departure Time	Arrival Time	Actual Arrival Time
0	da8a6ba8-b3dc-4677-b176	2023-12-08	12:41:11	Online	Contactless	Adult	Standard	Advance	43	London Paddington	Liverpool Lime Street	2024-01-01	11:00:00	13:30:00	13:30:00
1	b0cdd1b0-f214-4197-be53	2023-12-16	11:23:01	Station	Credit Card	Adult	Standard	Advance	23	London Kings Cross	York	2024-01-01	09:45:00	11:35:00	11:40:00
2	f3ba7a96-f713-40d9-9629	2023-12-19	19:51:27	Online	Credit Card	NaN	Standard	Advance	3	Liverpool Lime Street	Manchester Piccadilly	2024-01-02	18:15:00	18:45:00	18:45:00

From messy to clean

Missing values — handled with logic, not guesses

Column	Missing	Action
Railcard	20,918 (66%)	Filled with “No Railcard” — absence is meaningful
Reason for Delay	27,481 (87%)	On Time → “No Delay”; Delayed/Cancelled → “Unknown”
Actual Arrival Time	1,880 (6%)	Cancelled → “Cancelled”; else → “Unknown”

Type fixes

Date of Purchase &
Date of Journey → datetime64

Duplicates

0 fully duplicate rows found

Standardised

Weather Conditions→Weather,
Signal failure→Signal Failure,
Staff Shortage→Staffing

Interactive Excel dashboard

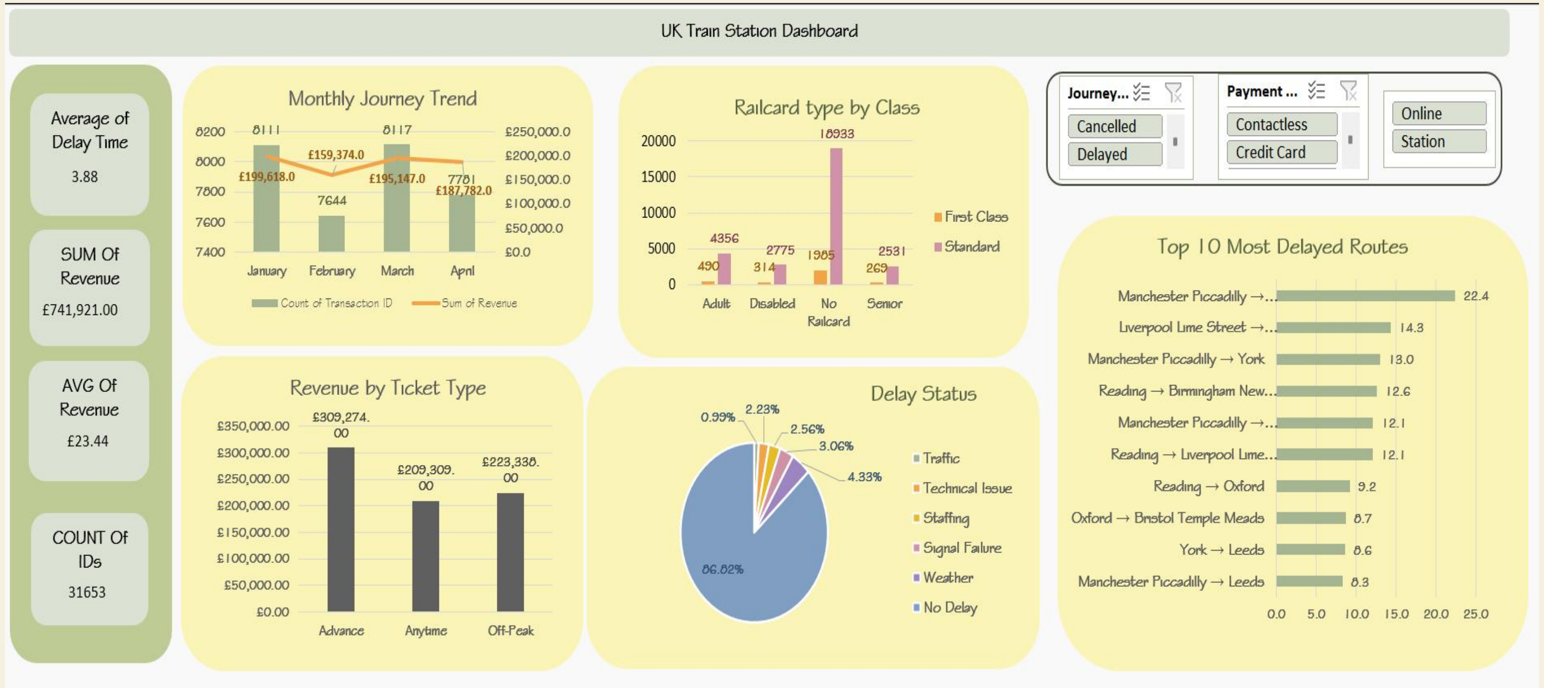
6 charts • 4 KPI cards • 3 connected slicers

3.88 min
AVG Delay Time

£741,921
Total Revenue

£23.44
AVG Ticket Price

31,653
Total Journeys



What each visual answers

Monthly Journey Trend

Combo chart — journeys (bars) + revenue (line), Jan–Apr 2024.

Revenue by Ticket Type

Advance leads at £309,274.

Railcard × Class

No-Railcard Standard dominates at 18,933.

Delay Status

Pie — 86.82% no delay.

Top 10 Delayed Routes

Manchester Piccadilly → London St Pancras at 22.4 min.

Slicers

Journey Status • Payment Method • Purchase Type.

Five queries, five answers

Data imported from Excel into a clean SQL Server .

Q1	On-Time Rate by Route
Q2	Avg & Max Delay by Reason
Q3	Monthly Revenue Trend
Q4	Refunds by Journey Status
Q5	Top 5 Most Delayed Routes

```sql

--- On-Time Rate by Route

SELECT
    Route,
    COUNT(*)                                                          AS Total_Journeys,
    SUM(CASE WHEN Journey_Status = 'On Time' THEN 1 ELSE 0 END)       AS On_Time_Count
FROM railway
GROUP BY Route
ORDER BY Total_Journeys DESC;
```

100 %   No issues found

Results   Messages

	Route	Total_Journeys	On_Time_Count
1	Manchester Piccadilly → Liverpool Lime Street	4628	3984
2	London Euston → Birmingham New Street	4209	3756
3	London Kings Cross → York	3922	3593
4	London Paddington → Reading	3873	3521
5	London St Pancras → Birmingham New Street	3471	3198
6	Liverpool Lime Street → Manchester Piccadilly	3002	2771
7	Liverpool Lime Street → London Euston	1097	218
8	London Euston → Manchester Piccadilly	712	667
9	Birmingham New Street → London St Pancras	702	651
10	London Paddington → Oxford	485	422
11	Manchester Piccadilly → London Euston	345	102
12	London St Pancras → Leicester	337	321
13	York → Durham	258	225

Top 5 most delayed routes

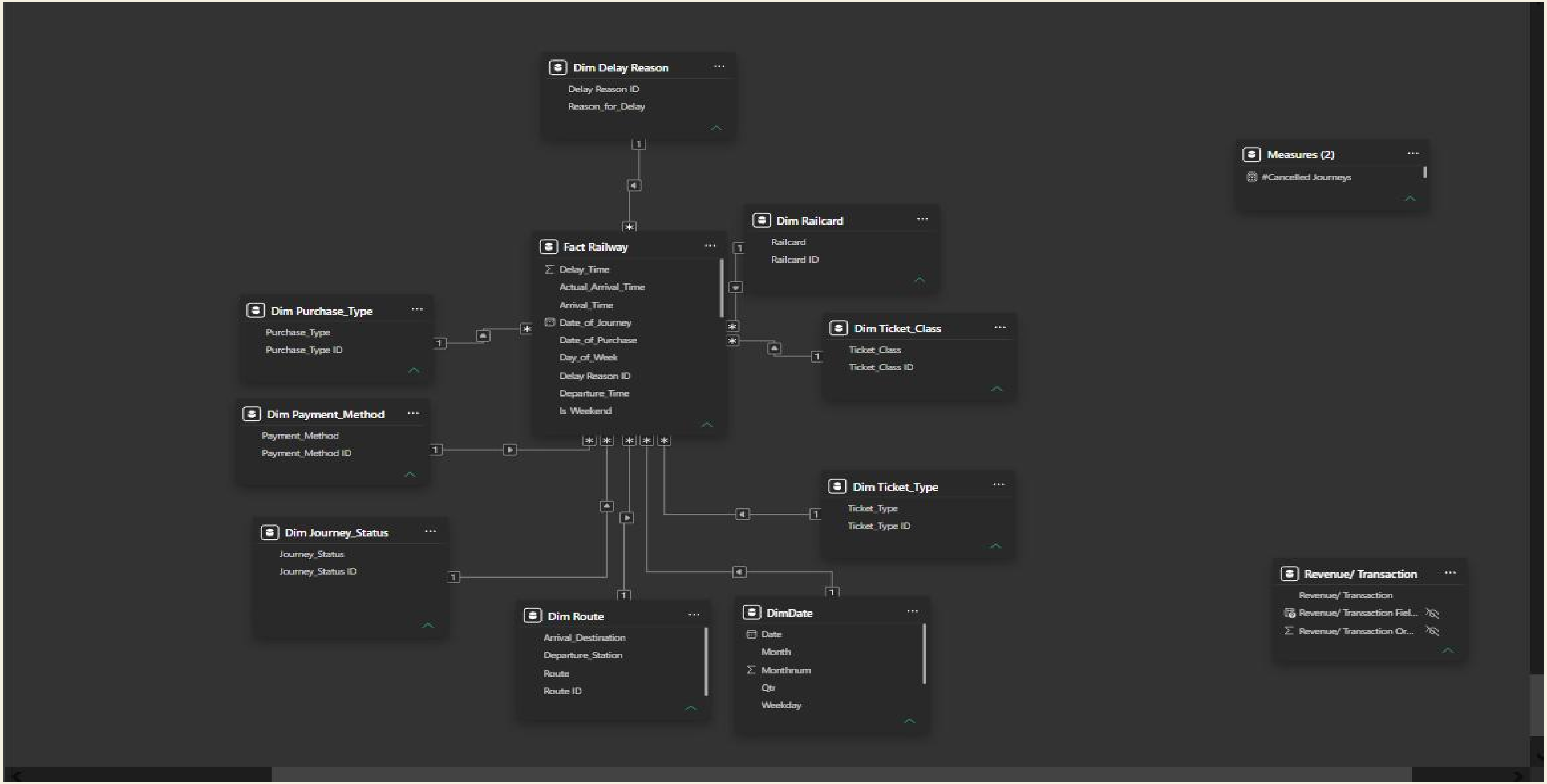
query_5.sql

```
SELECT TOP 5
    Route,
    COUNT(*)                AS Total_Delayed,
    ROUND(AVG([ Delay_Time]),2)  AS Avg_Delay_Mins
FROM Railway
WHERE Journey_Status = 'Delayed'
GROUP BY Route
ORDER BY Total_Delayed DESC;
```

INSIGHT

A handful of routes drive most of the delay volume — perfect targets for operational focus.

Combined with Q2 (delay reasons), these routes become a prioritised remediation backlog.



4 dimensions × 1 fact — efficient, intuitive filtering across every report page.

Overview · Revenue · Performance

01 OVERVIEW

KPIs

Transactions, AVG Delay, Net Revenue, AVG Ticket Price

INSIGHT

Credit Card leads payments (61.29%). Top crowded route: Manchester Piccadilly → Liverpool (1,110).

02 REVENUE

KPIs

Total Revenue, Expenses, Net Revenue, Refund Gauge

INSIGHT

Advance tickets highest at £309,274. Refund rate 3.47% — above 2% target.

03 PERFORMANCE

KPIs

On-Time, Delayed, Cancelled gauges + delay analysis

INSIGHT

On-time 87.03% (vs 90% target). Weather worst in April at 32.93 min.

Overview

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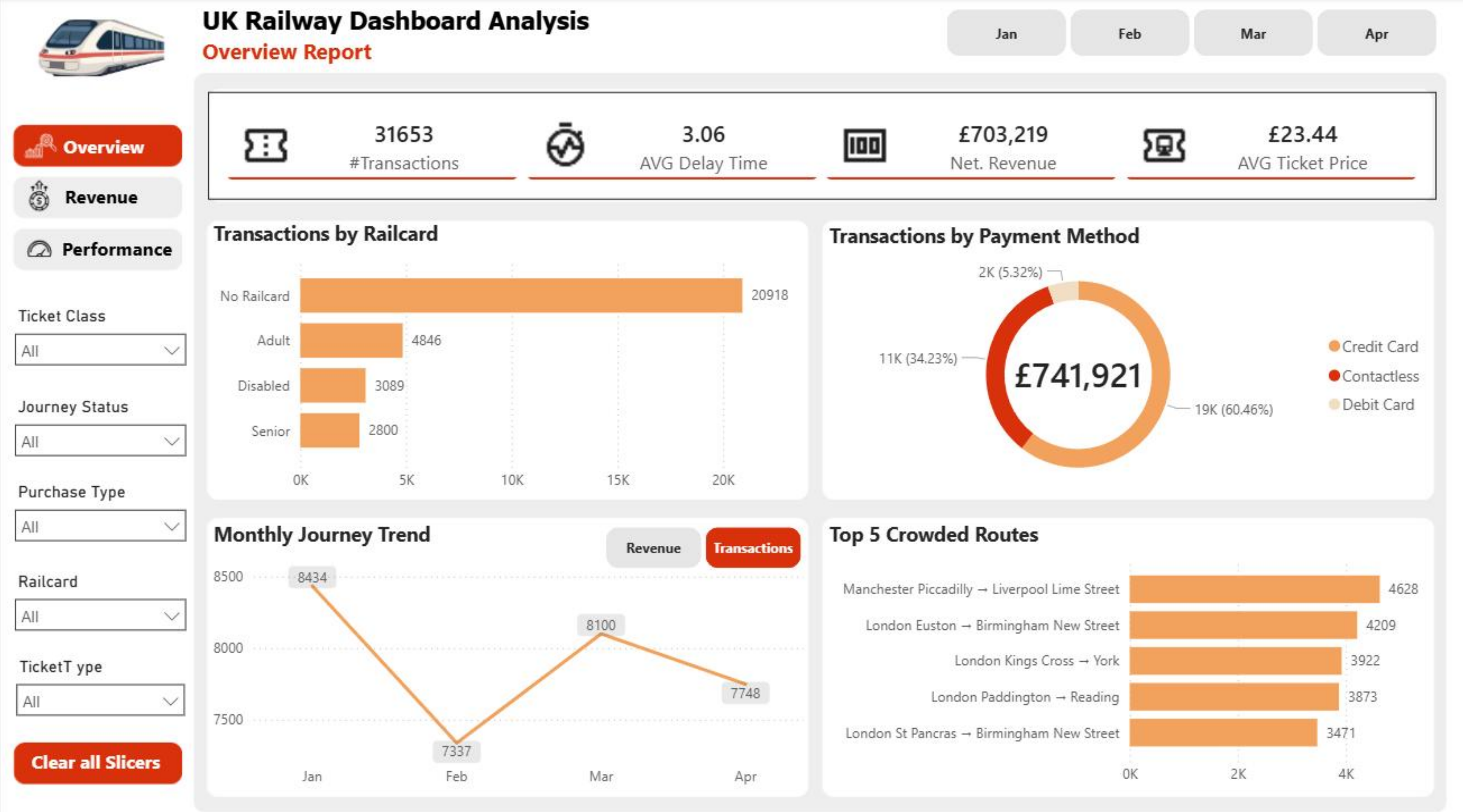

Revenue

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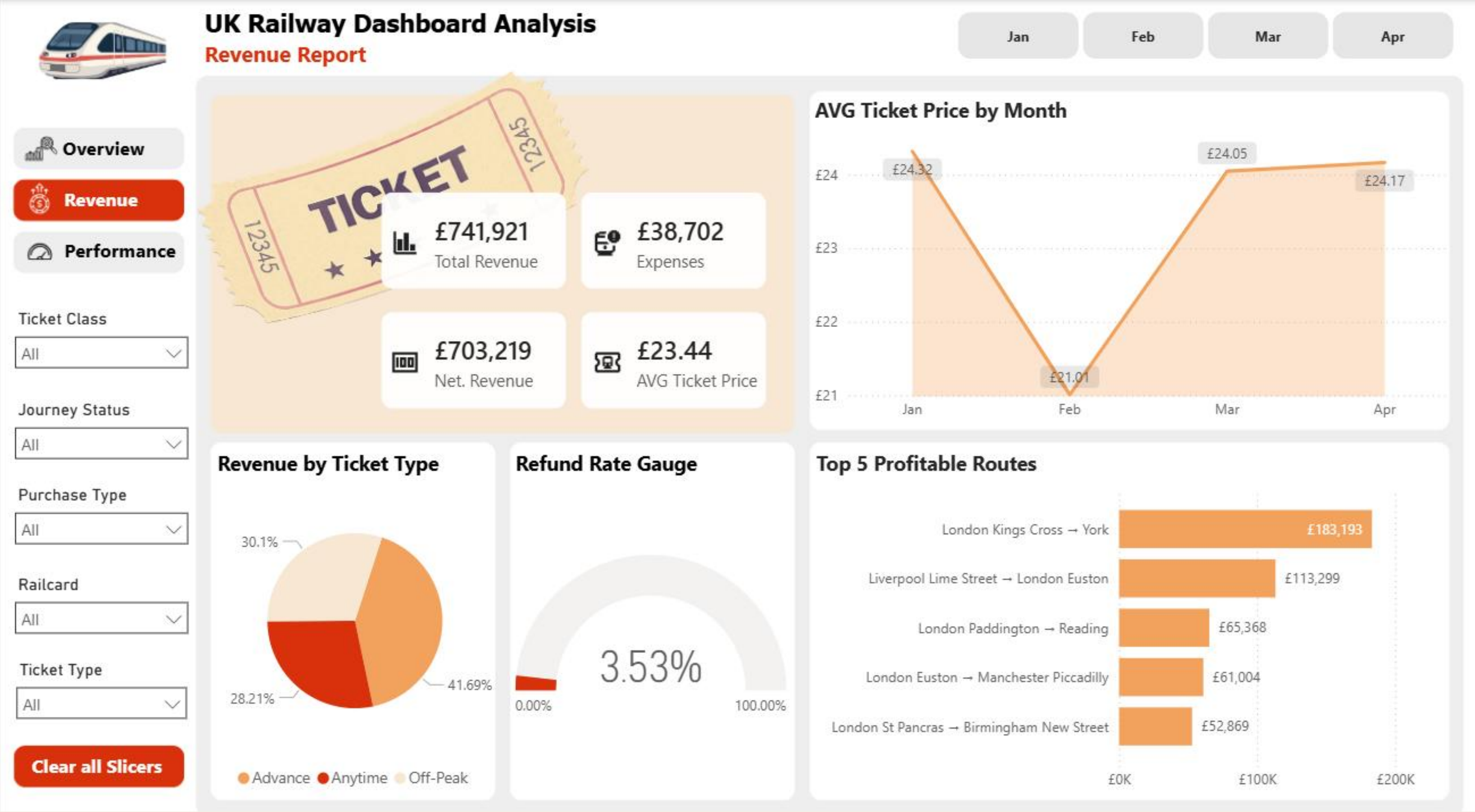

Performance

03 PERFORMANCE

KPIs

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INSIGHT

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How we measure success

KPI	Current	Target	Status
On-Time Rate	**87.03%**	90%	On target
Refund Rate	**3.47%**	Below 2%	Above
Cancelled Rate	**6.16%**	Below 3%	Above
AVG Delay Time	**3.88 min**	Below 5 min	On target

According to the Industry Benchmark for train stations: the on-Time rate should be from 85–90%"